

Chinook Salmon Freshwater Habitat Potential Modeling and Mapping for the Yukon and Kuskokwim River Basins in Alaska

Jeff Falke^{1,2,3} and Josh Paul^{2,3}

¹U.S. Geological Survey, ²Alaska Cooperative Fish and Wildlife Research Unit

³Institute of Arctic Biology, University of Alaska Fairbanks



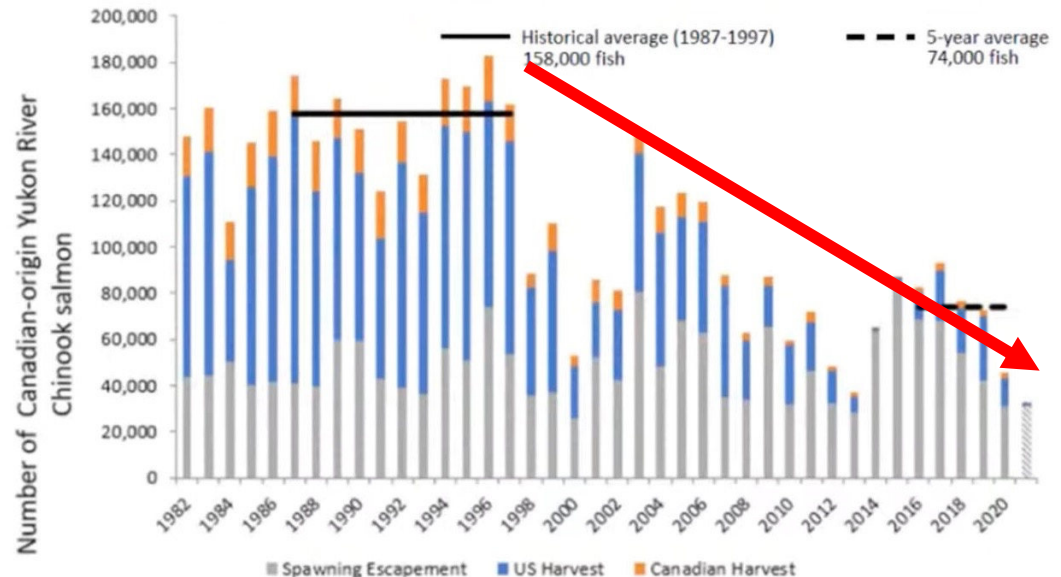
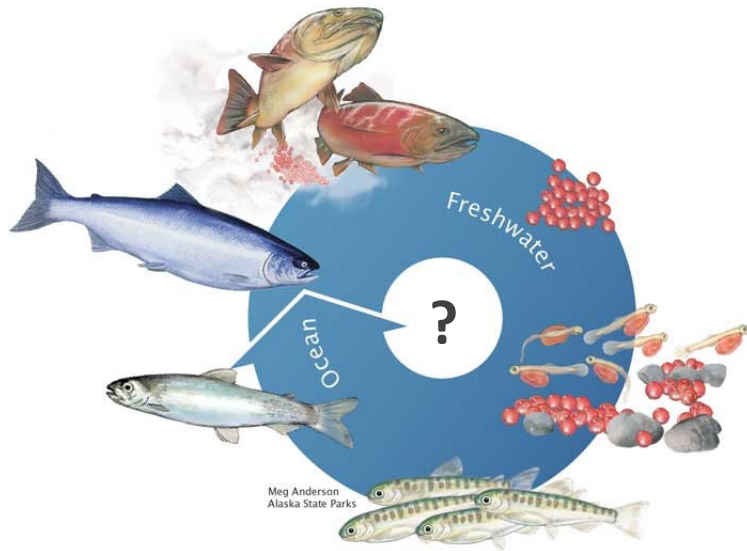
Outline

- Project Background and Justification
- Project Goals and Completion Status
- Methods: Describe Models and Mapping
- Methods: Case study example (Matter et al. 2018)
- Results: Yukon-Kuskokwim Habitat Potential
- Potential Applications: Examples from interior AK
- Decision Support and Ongoing Research

Project Background and Justification

Chinook Salmon

- Important commercial, cultural, subsistence, recreational, and ecological resource in Alaska
- Substantial population declines since 2000
- Lack of information on freshwater habitats



Ecological Applications, 17(1), 2007, pp. 66–80
© 2007 by the Ecological Society of America

DISTRIBUTION OF SALMON-HABITAT POTENTIAL RELATIVE TO LANDSCAPE CHARACTERISTICS AND IMPLICATIONS FOR CONSERVATION

KELLY M. BURNETT,^{1,4} GORDON H. REEVES,¹ DANIEL J. MILLER,² SHARON CLARKE,^{3,5} KEN VANCE-BORLAND,³
AND KELLY CHRISTIANSEN¹

¹*USDA Forest Service, Pacific Northwest Research Station, 3200 SW Jefferson Way, Corvallis, Oregon 97331 USA*

²*Earth Systems Institute, 3040 NW 57th Street, Seattle, Washington 98107 USA*

³*Department of Forest Science, Oregon State University, 321 Richardson Hall, Corvallis, Oregon 97331-5752 USA*

The metric, termed **intrinsic potential (IP)**, reflects species-specific associations between fish use and persistent stream attributes.

A relative ranking of fish habitat quality

Freshwater Habitat Potential for Chinook Salmon in the Yukon and Kuskokwim River Basins, Alaska

Project Goals

- ✓ 1. Develop synthetic stream networks for Y-K basins
- ✓ 2. Compile georeferenced juvenile & adult Chinook salmon distribution across Y-K basins
- ✓ 3. Produce maps of spawning & rearing habitat potential from modeled geomorphic attributes
- 4. Hold targeted stakeholder meetings to refine initial maps & models
- 5. Develop a spatially explicit decision support tool

Freshwater Habitat Potential for Chinook Salmon in the Yukon and Kuskokwim River Basins, Alaska

Project Goals

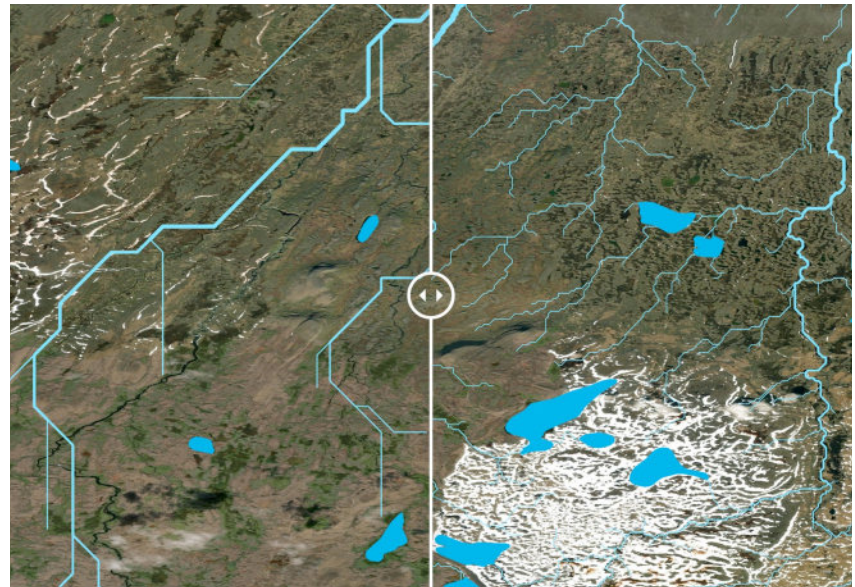
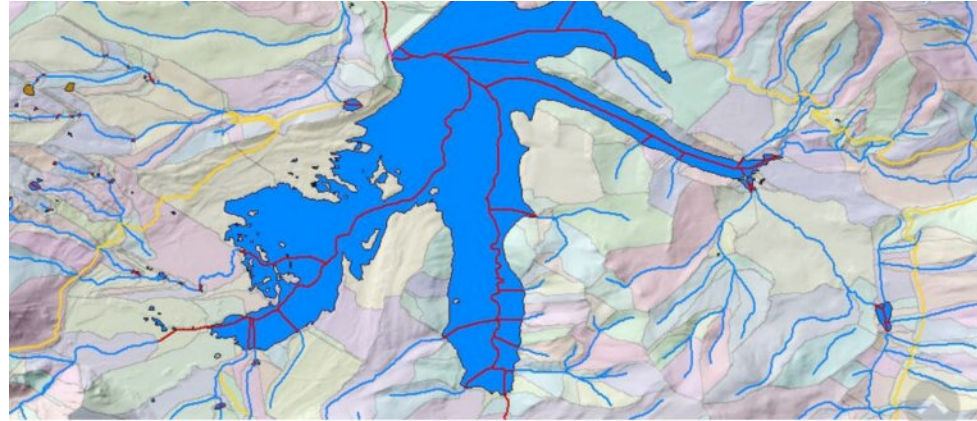
- 1. Develop synthetic stream networks for Y-K basins**
2. Compile georeferenced juvenile & adult Chinook salmon distribution across Y-K basins
3. Produce maps of spawning & rearing habitat potential from modeled geomorphic attributes
4. Hold targeted stakeholder meetings to refine initial maps & models
5. Develop a spatially explicit decision support tool

Where are the streams?

National Hydrography Dataset (Alaska)

- Coarse resolution 1:63,650
- Misrepresented flow lines
- Disconnected and omitted streams
- No attributes

HydroSheds v2 example (90 m² vs 12 m²)



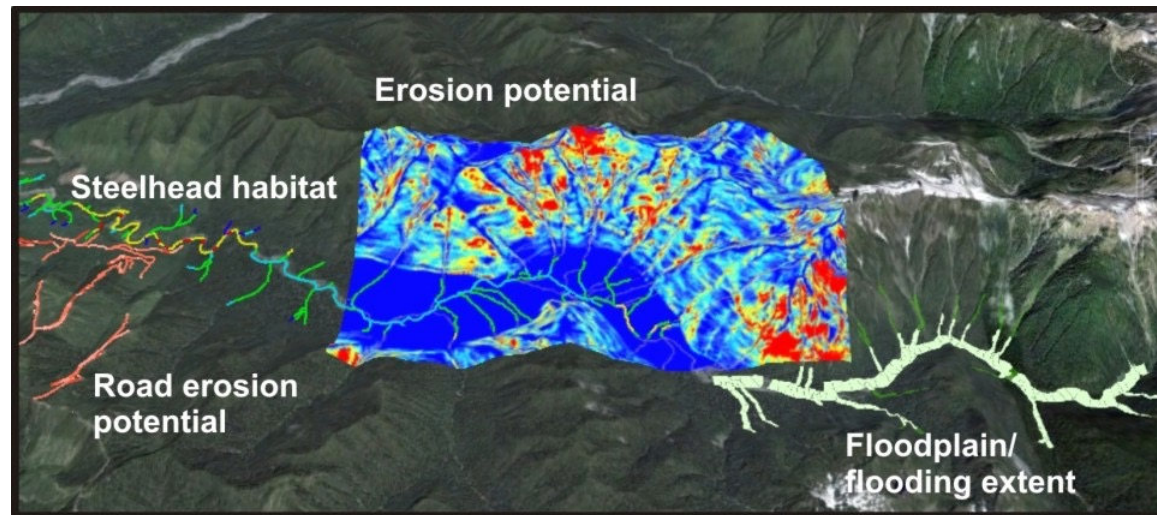
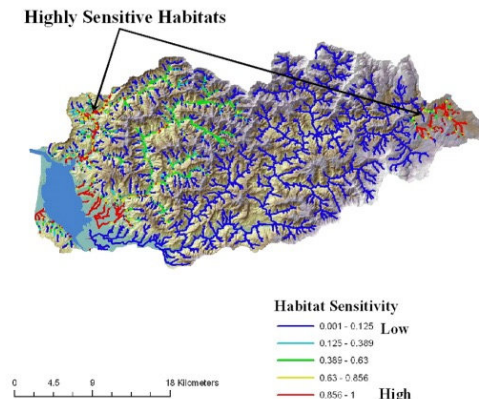
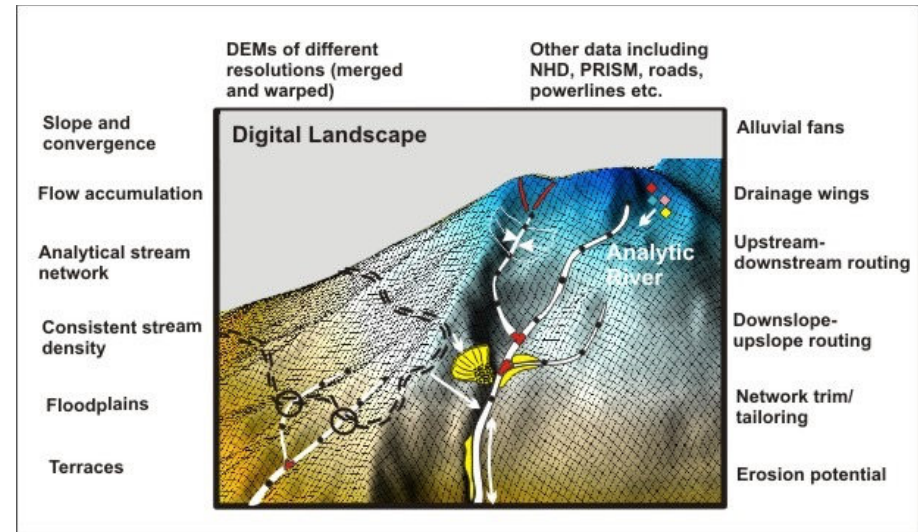
Synthetic Stream Networks “NetMap”

NetMap (TerrainWorks)

A system of “digital landscapes” for conducting environmental assessments
(Benda et al. 2007 *Forest Science*)

Capabilities include...

- Mapping aquatic habitats
- Locating biological hotspots
- Identifying areas for restoration
- Quantifying habitat-stressor interactions
- Isolating areas of high erosion potential



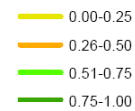
NetMap Coverage in Alaska + Applications

1. Tongass NF ([AK SeaGrant](#))
2. Copper River Basin ([Moore Foundation](#))
3. Nome River ([USFS](#))
4. Colville River ([EPSCoR](#); [BLM](#); [USGS](#))
5. Interior Alaska ([ADFG](#); [State of Alaska](#); [DoD](#))
6. Matanuska-Susitna River Basins ([ADFG](#); [USGS](#))
7. Yukon-Kuskokwim ([DoD](#); [USFWS](#))

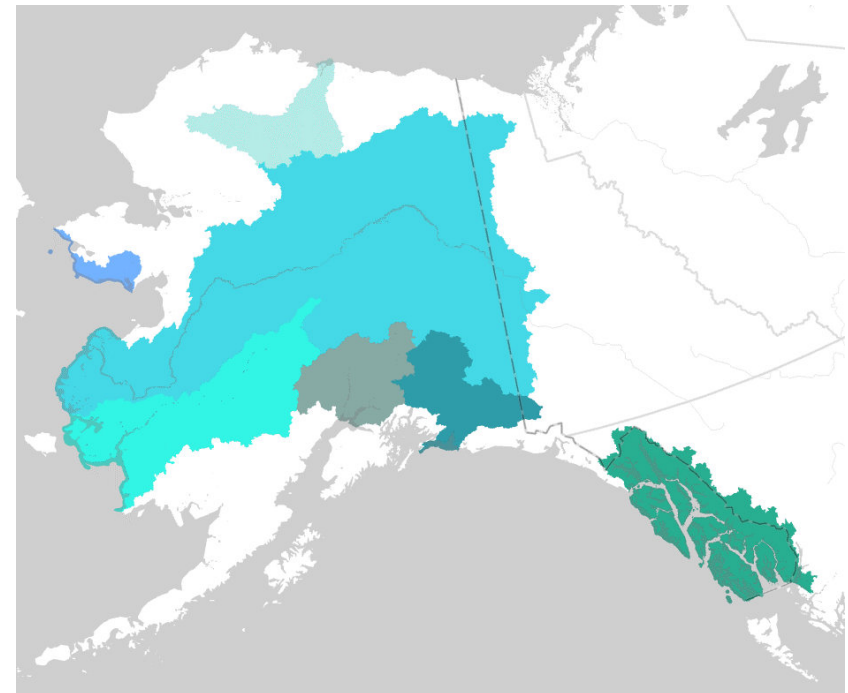
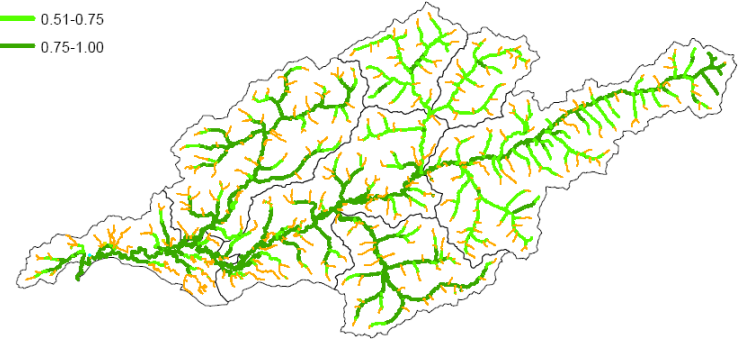
Applications

- Chinook salmon habitat potential (spawning and rearing) **Bidlack et al. 2014, Matter et al. 2018, Huntsman et al. 2017**
- Broad whitefish spawning habitats **Leppi et al. 2022**
- Potamodromous rainbow trout habitat selection **Fraley et al. 2016**
- Salmon vulnerability to Northern pike invasions **Jalbert et al. 2021**
- SE AK salmon life cycle modeling, hypoxia vulnerability **Sergeant 2022**

Habitat intrinsic potential

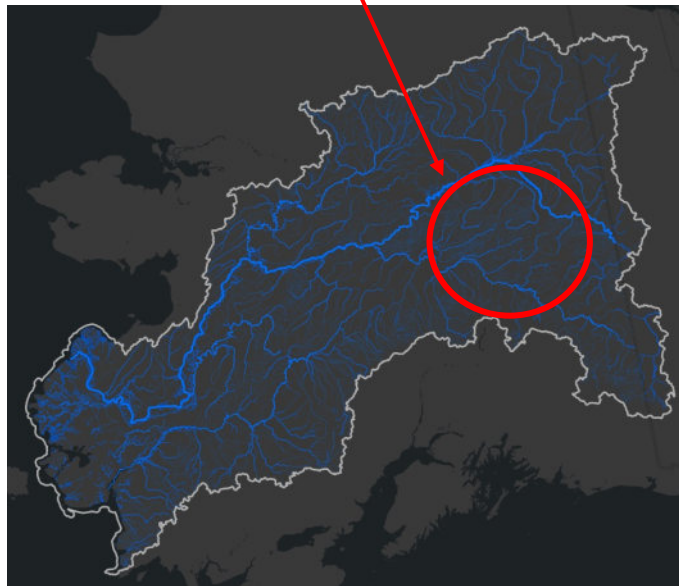
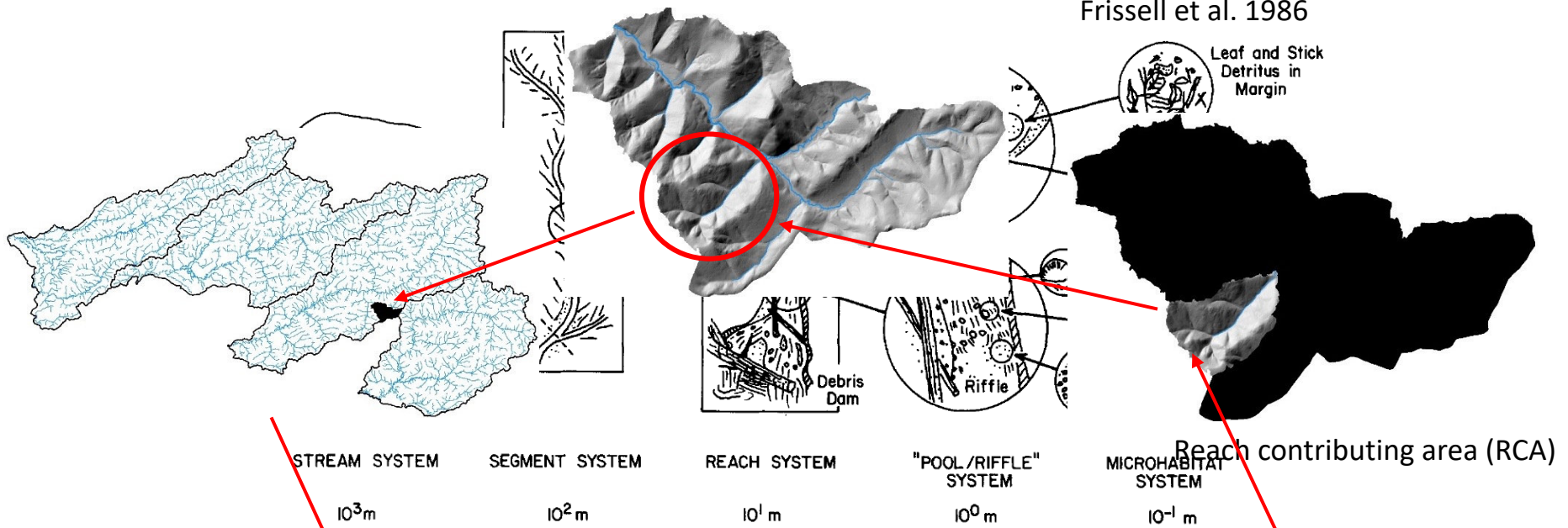


Matter et al. 2018

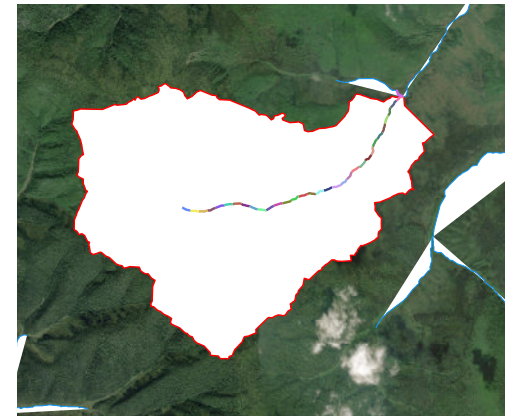


Data Structure

Frissell et al. 1986



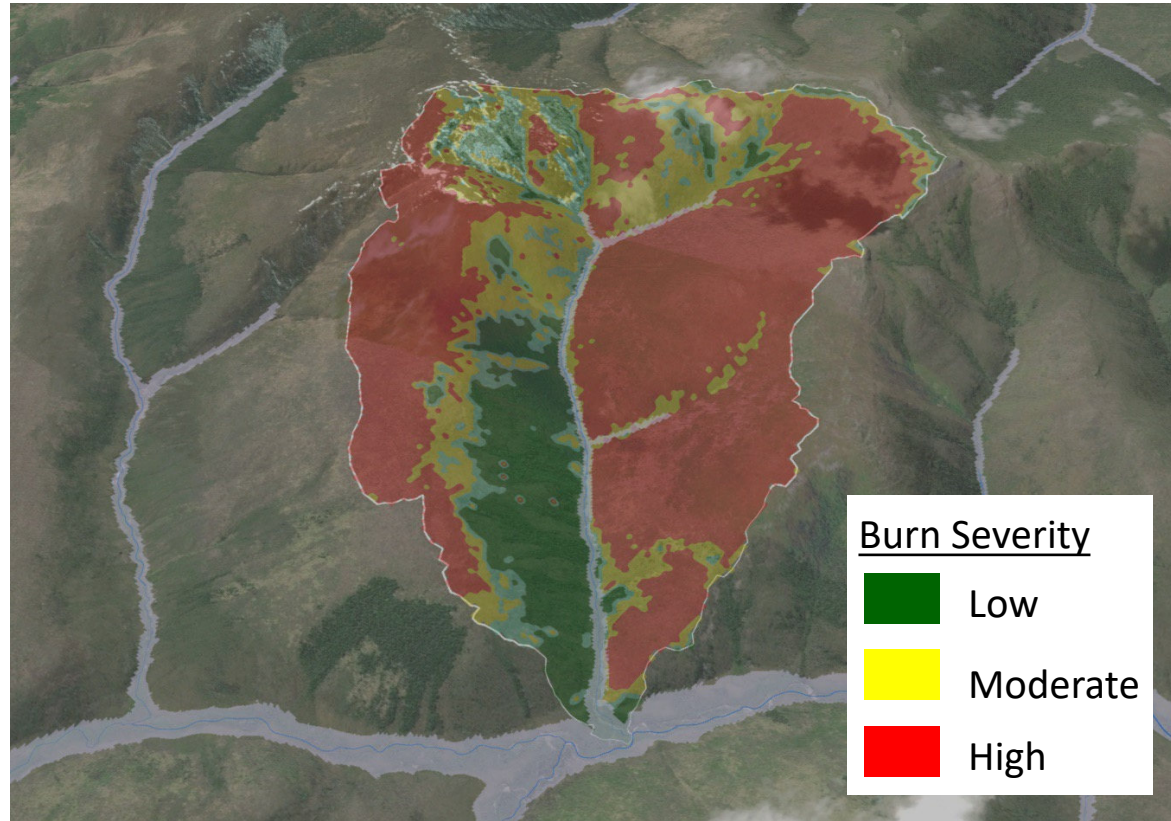
NetMap reach (50-100m)
Valley Segment (~1 km)
*2km² vs. 5km²



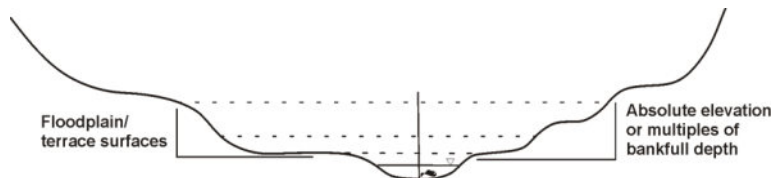
Floodplain and Valley Bottom Delineation

Components

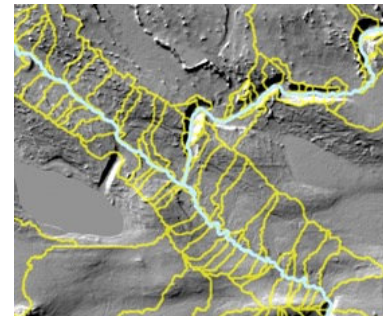
- Sub-catchment
- Streams
- Riparian Valley Bottom (RVB)
- Hillslopes



Riparian Valley Bottom (RVB)



NetMap (Benda et al. 2007)

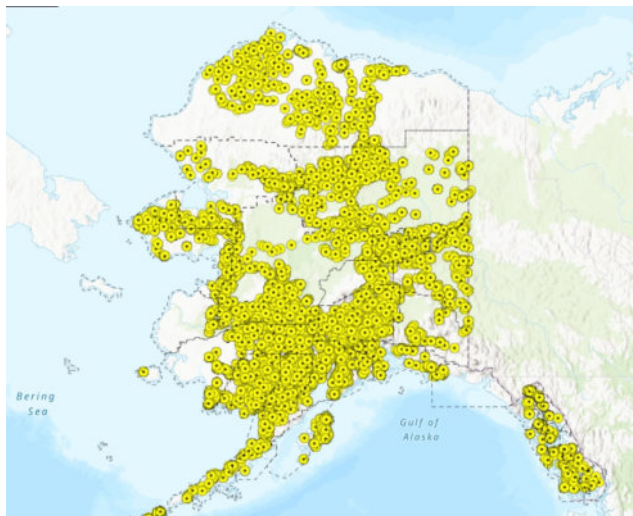


Freshwater Habitat Potential for Chinook Salmon in the Yukon and Kuskokwim River Basins, Alaska

Project Goals

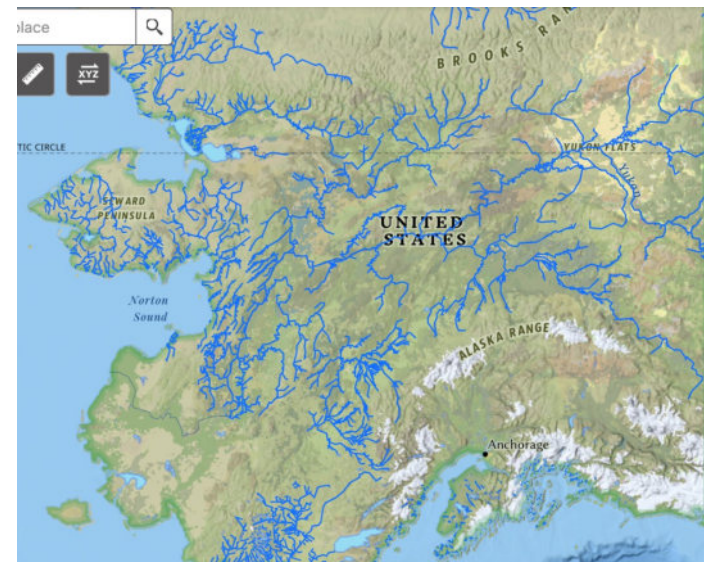
1. Develop synthetic stream networks for Y-K basins
- 2. Compile georeferenced juvenile & adult Chinook salmon distribution across Y-K basins**
3. Produce maps of spawning & rearing habitat potential from modeled geomorphic attributes
4. Hold targeted stakeholder meetings to refine initial maps & models
5. Develop a spatially explicit decision support tool

Where are the fish???



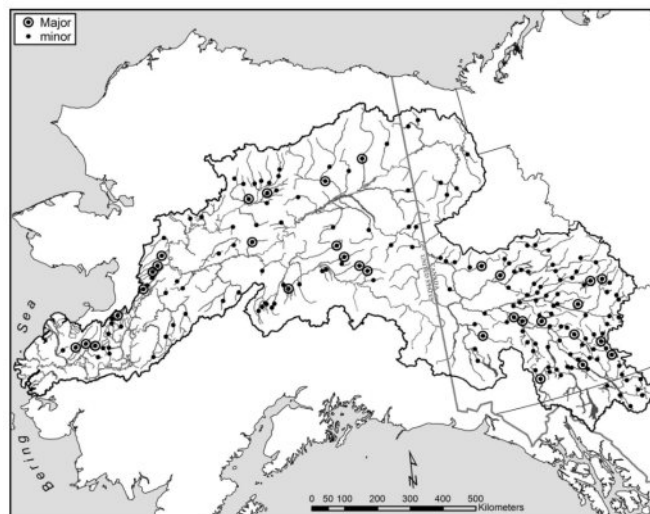
Alaska Freshwater Fish Inventory (AFFI)

- ADFG
- Juveniles
- Presence/absence
- Point Data



Yukon Chinook Spawning Catalog (Brown et al. 2017)

- USFWS
- Adults
- Presence/Abundance
- Point



Anadromous Waters Catalog (AWC)

- ADFG
- Adults and juveniles
- Presence*
- Polyline/Point

Other sources?

ADFG Comm. Fish. Counts

Schoen/Rinella project

???

Freshwater Habitat Potential for Chinook Salmon in the Yukon and Kuskokwim River Basins, Alaska

Project Goals

1. Develop synthetic stream networks for Y-K basins
2. Compile georeferenced juvenile & adult Chinook salmon distribution across Y-K basins
- 3. Produce maps of spawning & rearing habitat potential from modeled geomorphic attributes**
4. Hold targeted stakeholder meetings to refine initial maps & models
5. Develop a spatially explicit decision support tool

Diverse Habitats



Rearing Habitat Potential

Geomorphic attributes

Reach gradient (%)

- Energetics / velocity barriers

Mean annual discharge (m³/s)

- Proxy for stream size

Valley constraint (ratio; unitless)

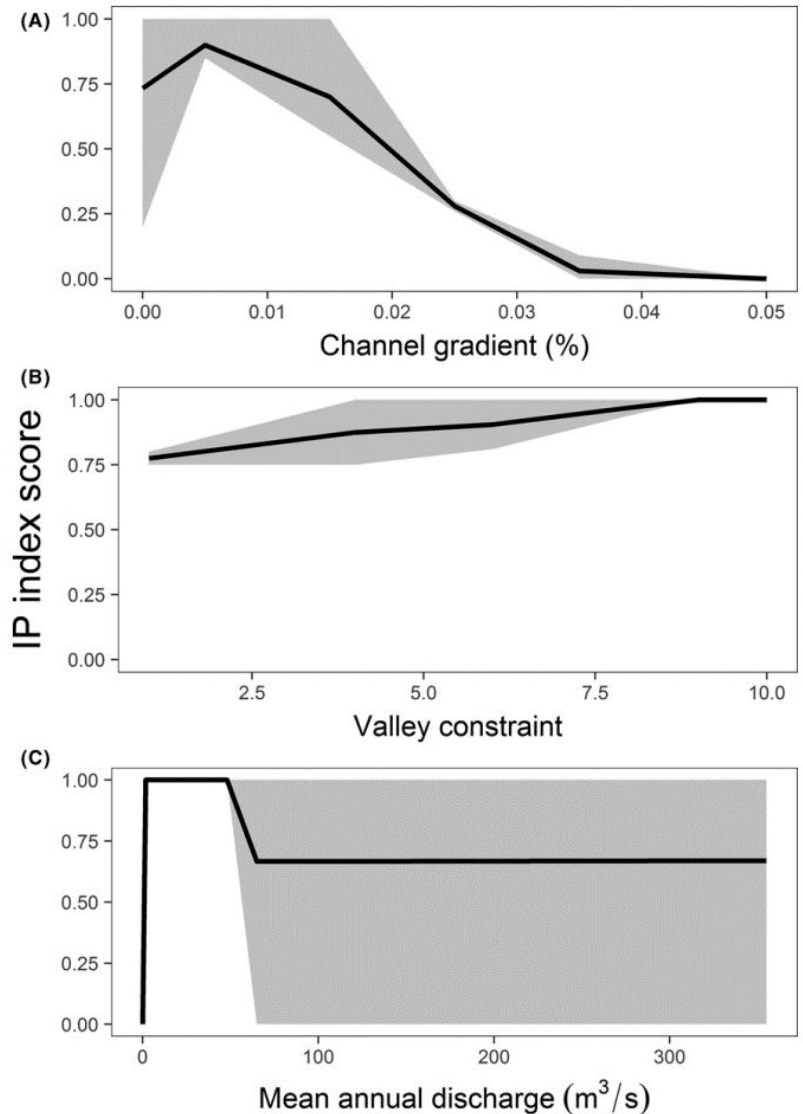
- Habitat complexity
- Large wood accumulation?

Intrinsic Habitat Potential

(Burnett et al. 2007 *Ecol. Appl.*)

$$IP = (V_1 * V_2 * V_3 \dots V_n)^{1/n}$$

Matter (2016); Matter et al. 2018

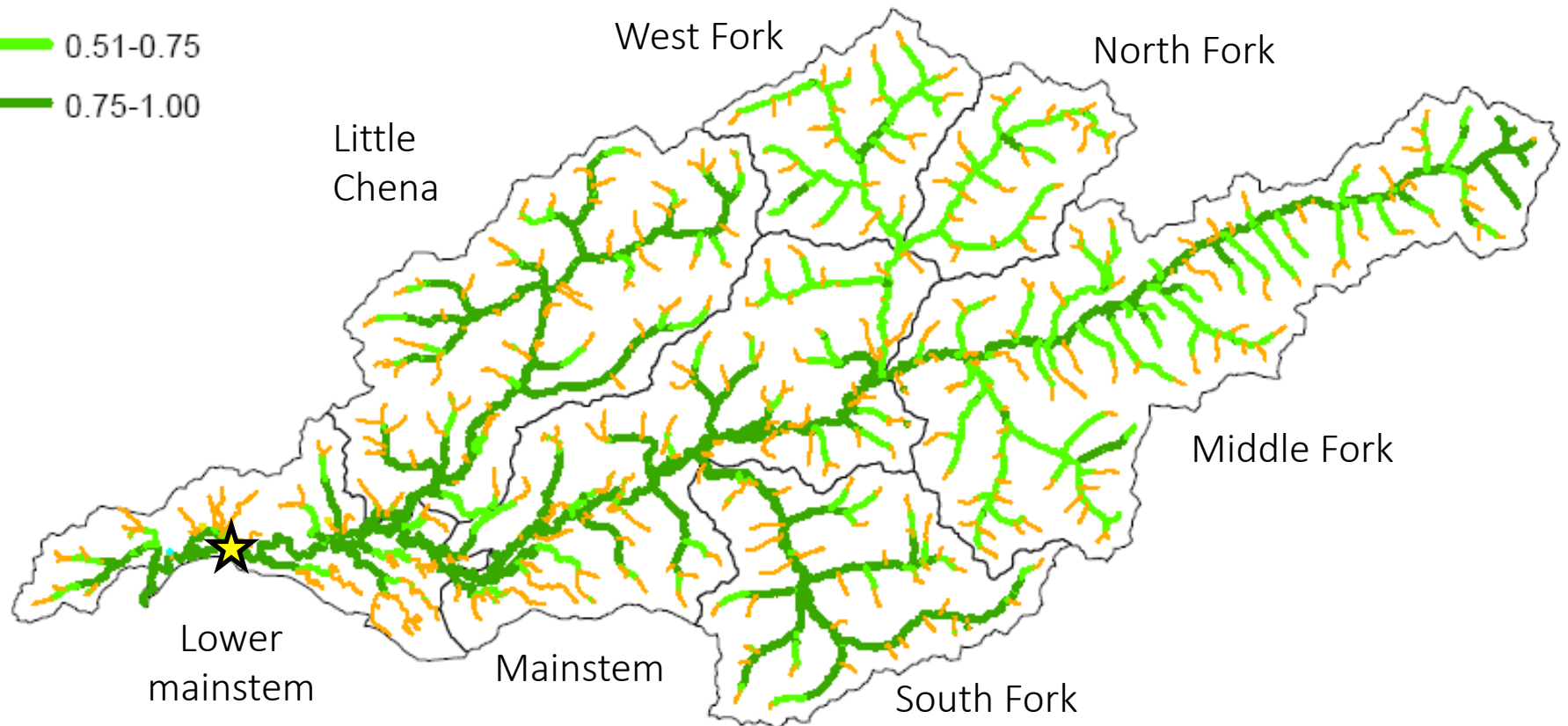
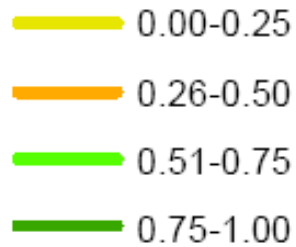


Rearing Habitat Potential

Matter (2016); Matter et al. 2018

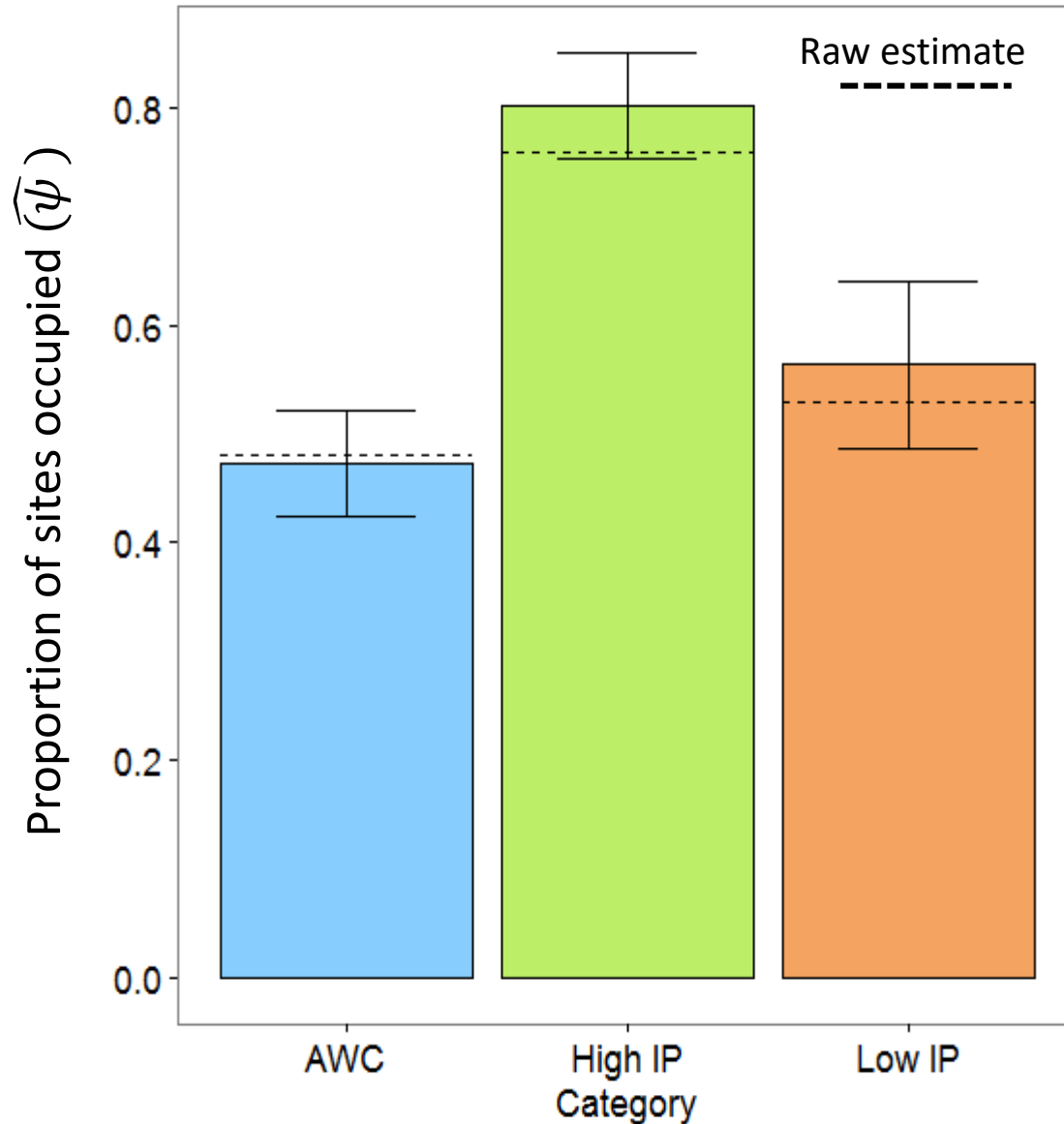
- 2,265 total stream-km
- 931 stream-km, $IP \geq 0.75$

Habitat intrinsic potential



Rearing Habitat Potential – eDNA Validation

Matter (2016); Matter et al. 2018



Y-K Habitat Potential

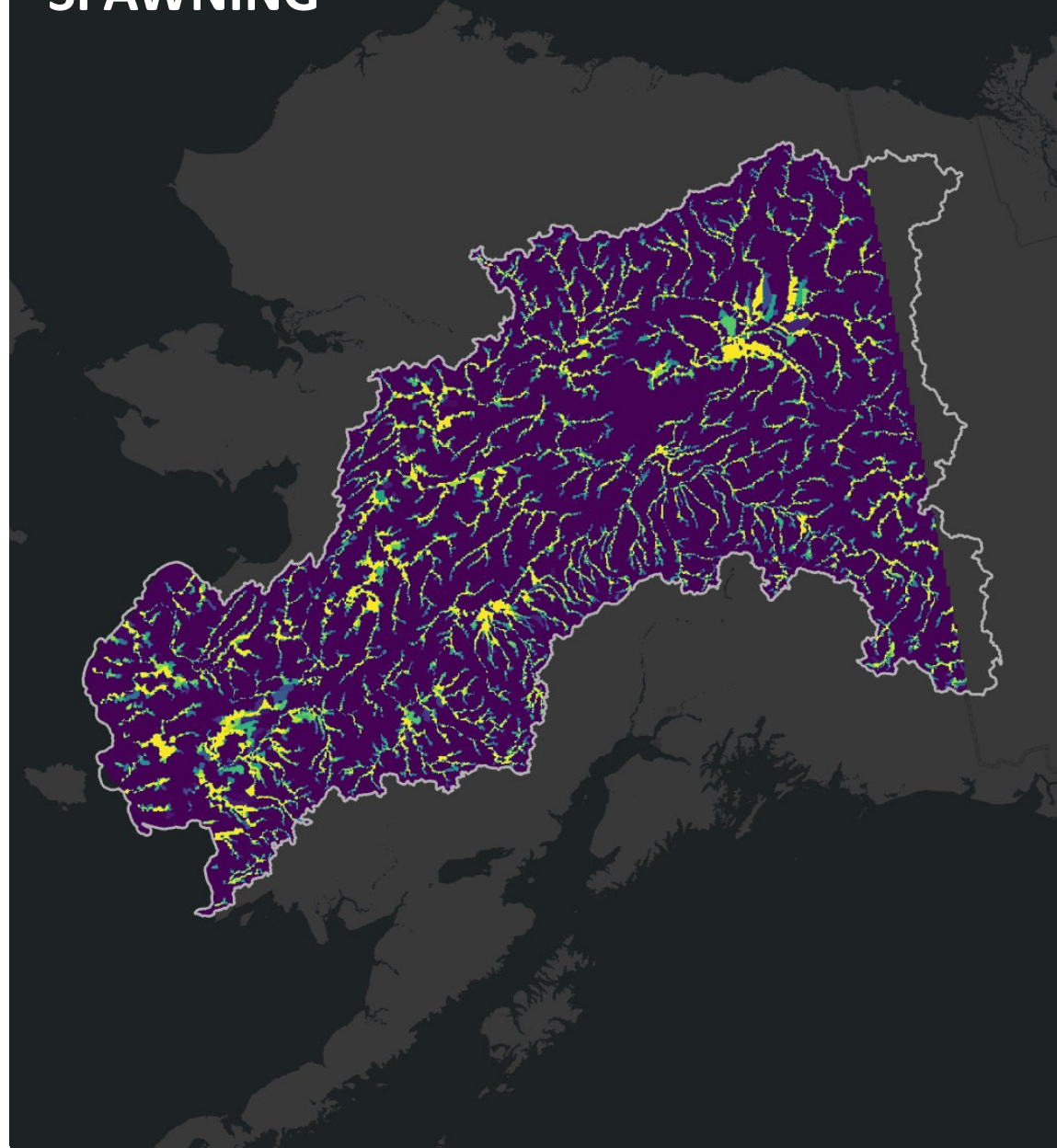
Y-K Summary & Overview

- Synthetic Stream Network generated for entire Y-K basin
 - ~667,000 km of streams
 - >1.3 million km²
- Rearing habitat IP model applied
Matter et al. 2018
 - 64,222 km JIP > 0.75
 - 87,542 km JIP > 0.6
- Spawning habitat IP model applied
Busch et al. 2011
 - 26,773 km SIP > 0.75
 - 39,446 km SIP > 0.6

Habitat intrinsic potential

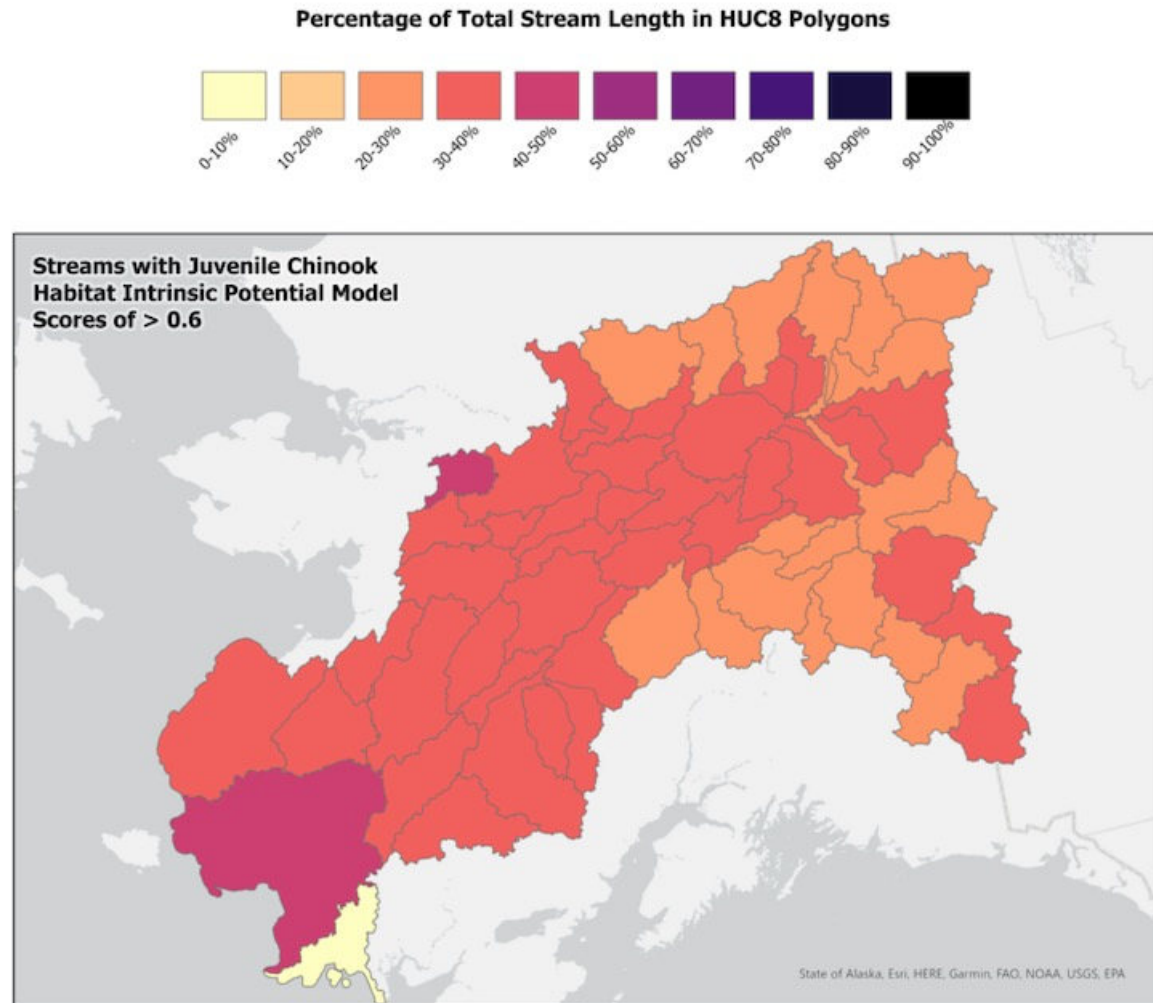


SPAWNING



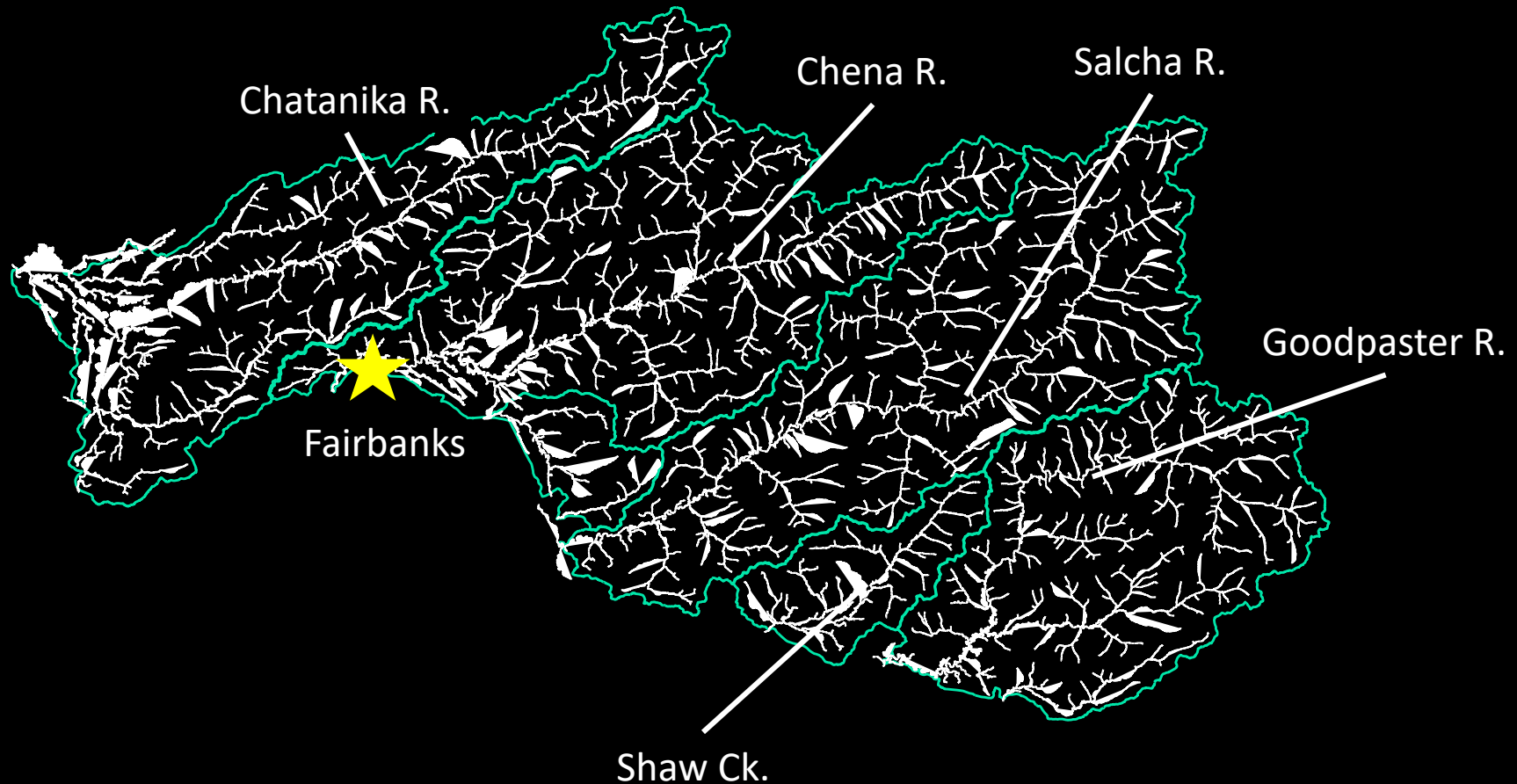
Y-K Habitat Potential: Applications

- What is the overlap between “high” IP and the known distribution of fish?
- What percentage of stream length in a basin is “high” IP?
- What is the relative distribution of habitats with “high” rearing *and* spawning habitats?
- What “high” IP locations are outside the AWC?

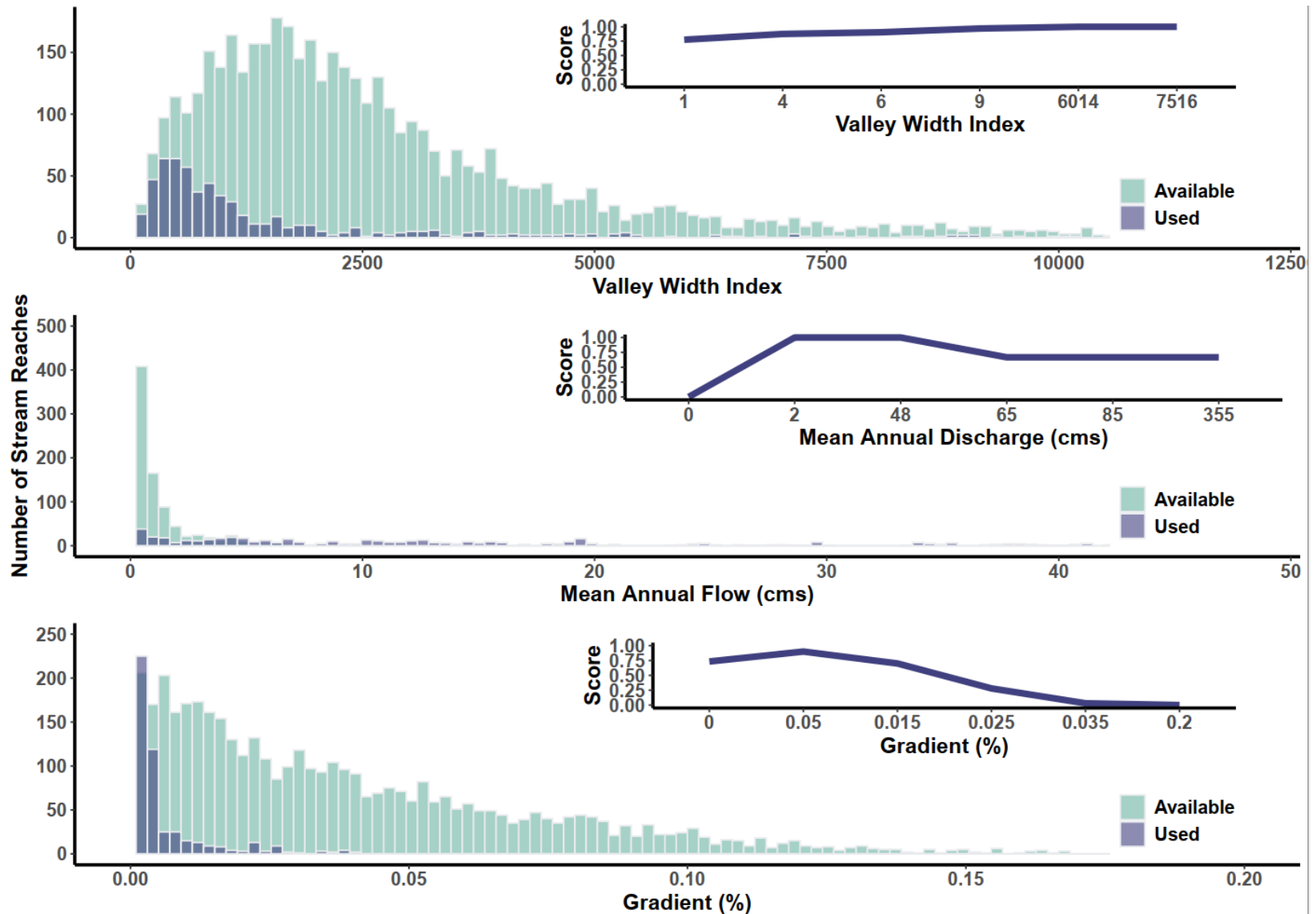


Y-K Habitat Potential: Applications

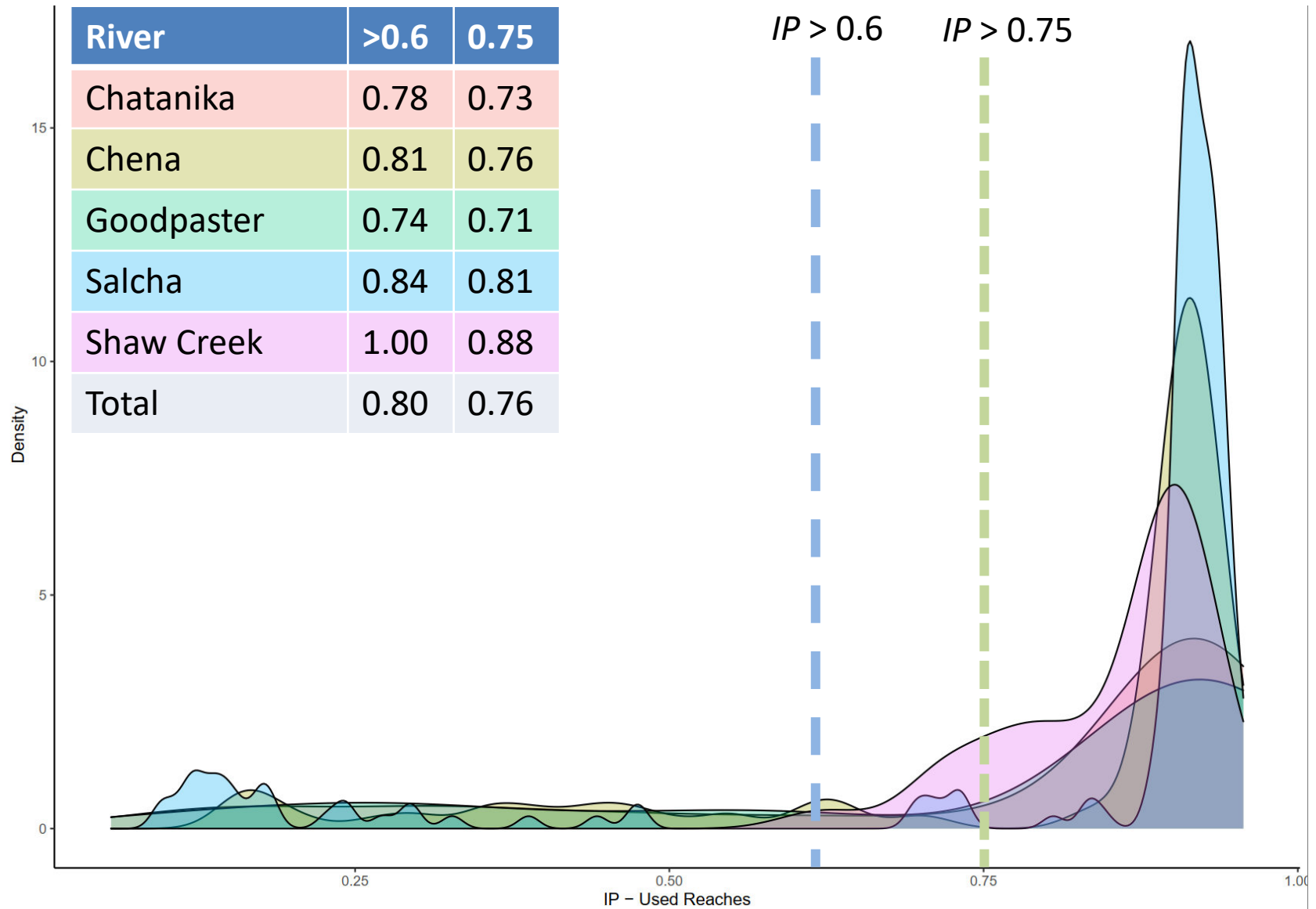
What is the overlap between “high” IP and the known distribution of fish?



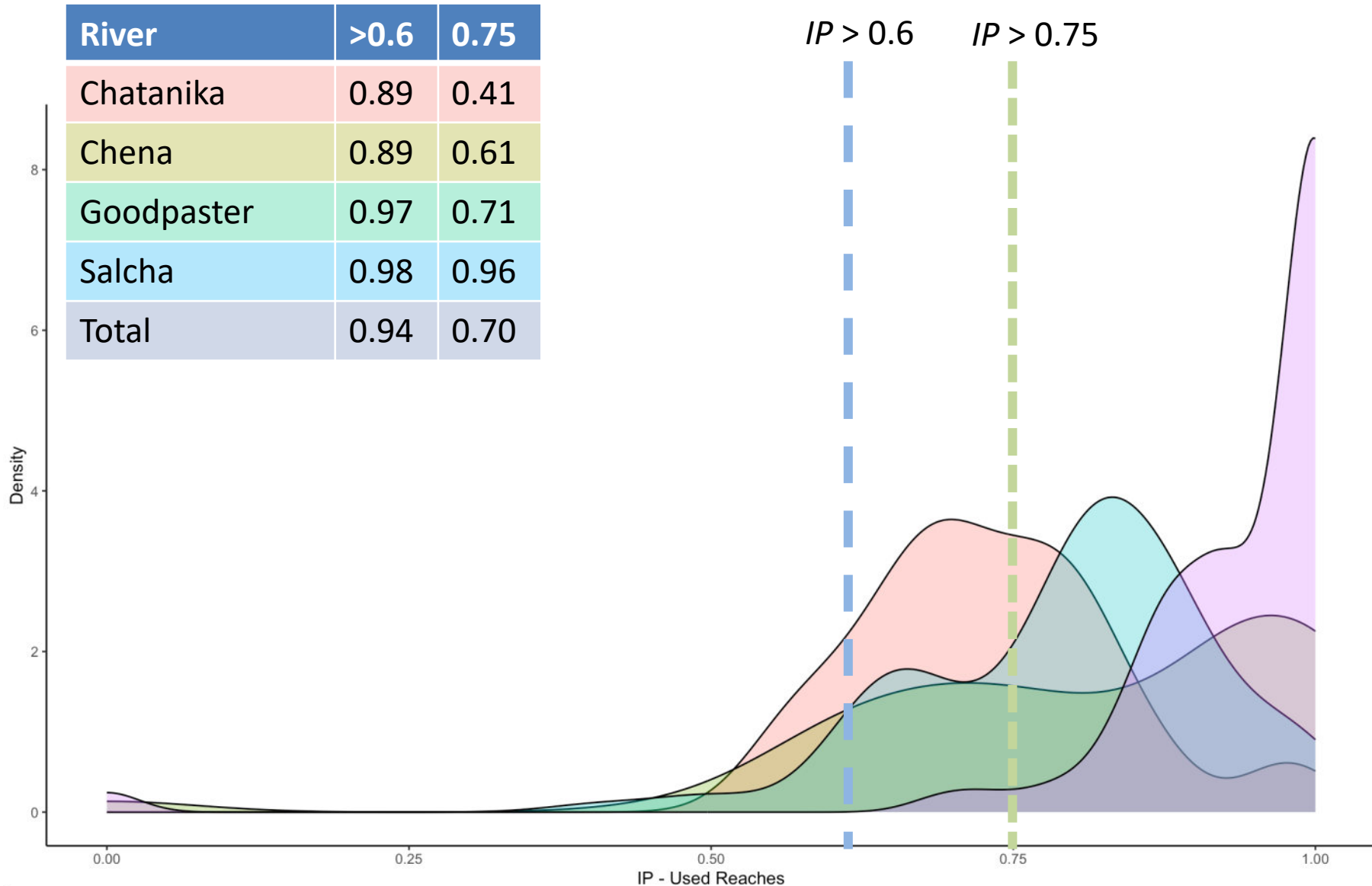
Uncertainty: Rearing Habitat Potential (Interior AK)



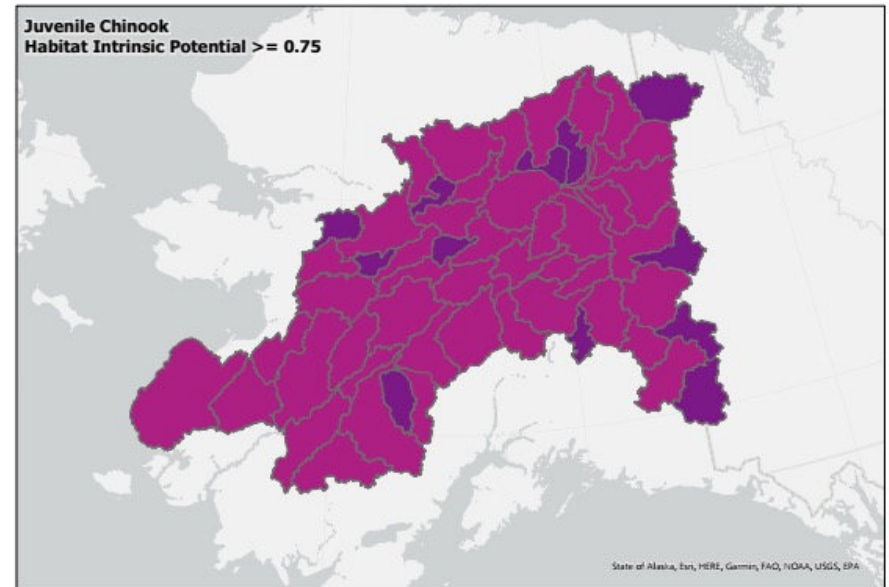
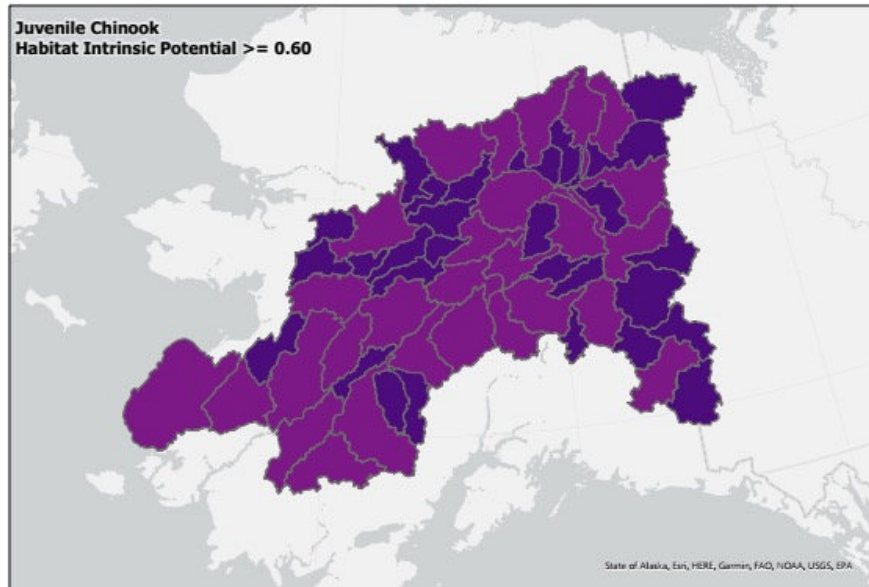
Uncertainty: Rearing Habitat Potential (Interior AK)



Uncertainty: Spawning Habitat Potential (Interior AK)



Uncertainty: Rearing Habitat Potential (Y-K)



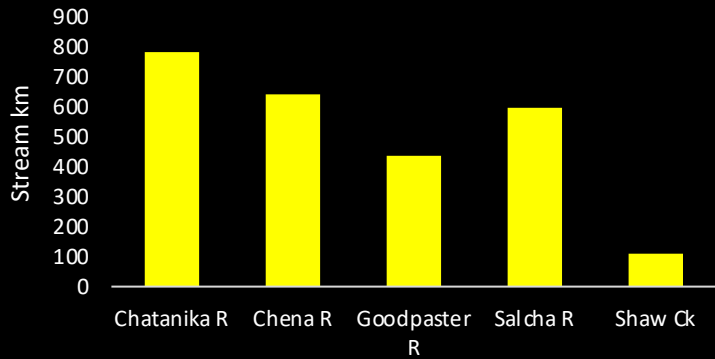
**Percentage of Area Under Cumulative Distribution Curve:
Stream Reaches Used by Juvenile Chinook
Summarized by NHD HUC8 Polygons**



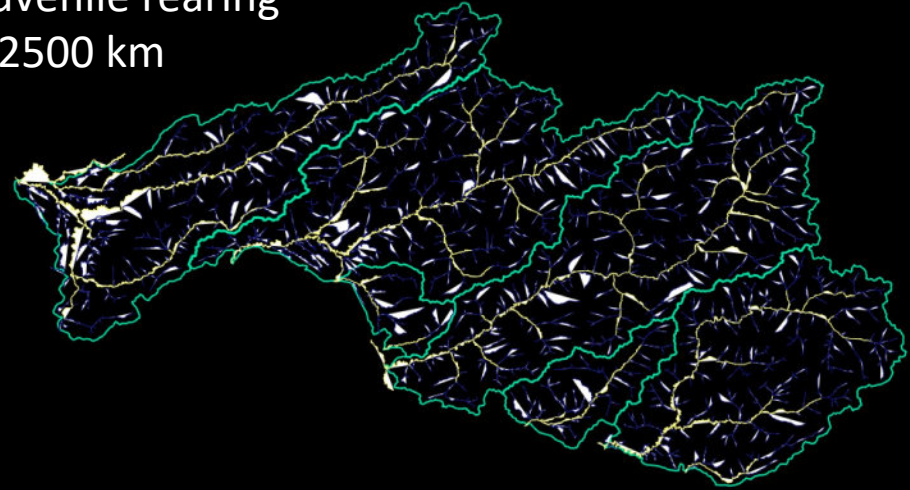
Y-K Habitat Potential: Applications

What percentage of stream length in a basin is “high” IP (i.e., > 0.6)?

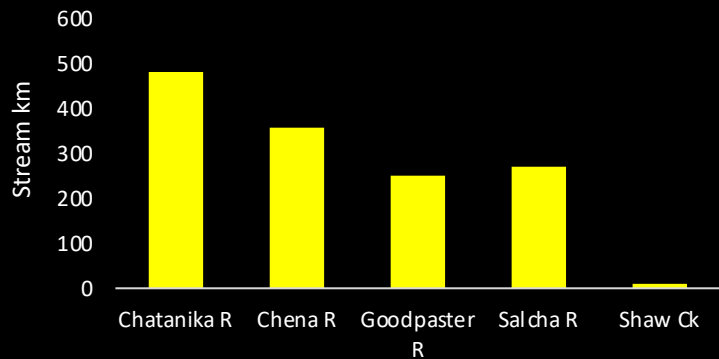
High rearing habitat potential



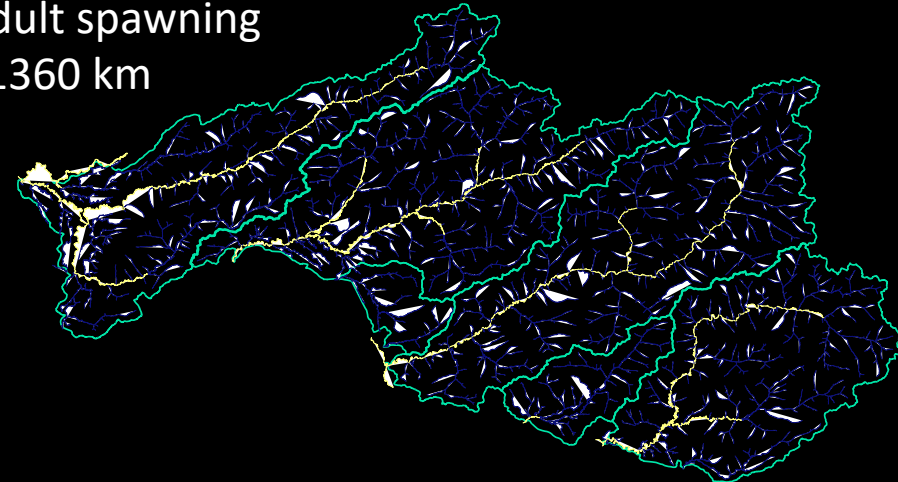
Juvenile rearing
~2500 km



High spawning habitat potential

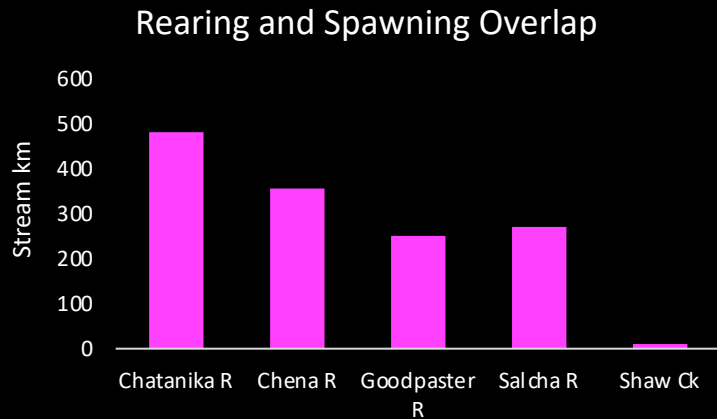


Adult spawning
~1360 km

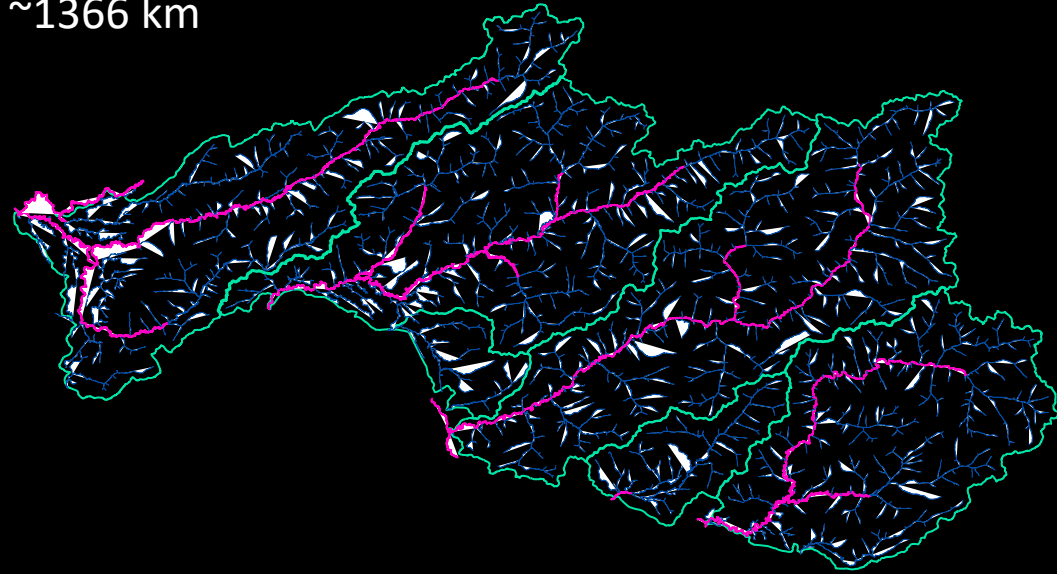


Y-K Habitat Potential: Applications

What is the relative distribution of habitats with “high” rearing *and* spawning habitat IP?



Juvenile rearing | spawning
~1366 km

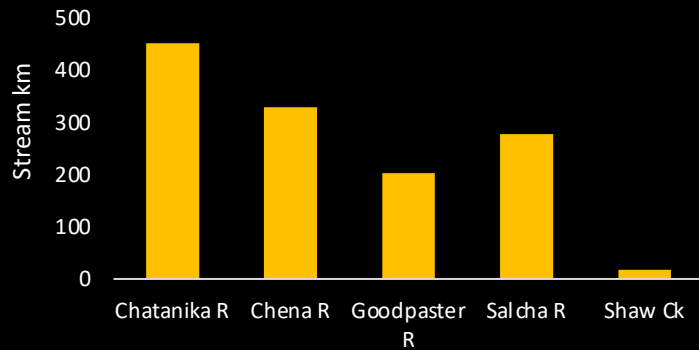


Interior Alaska basins

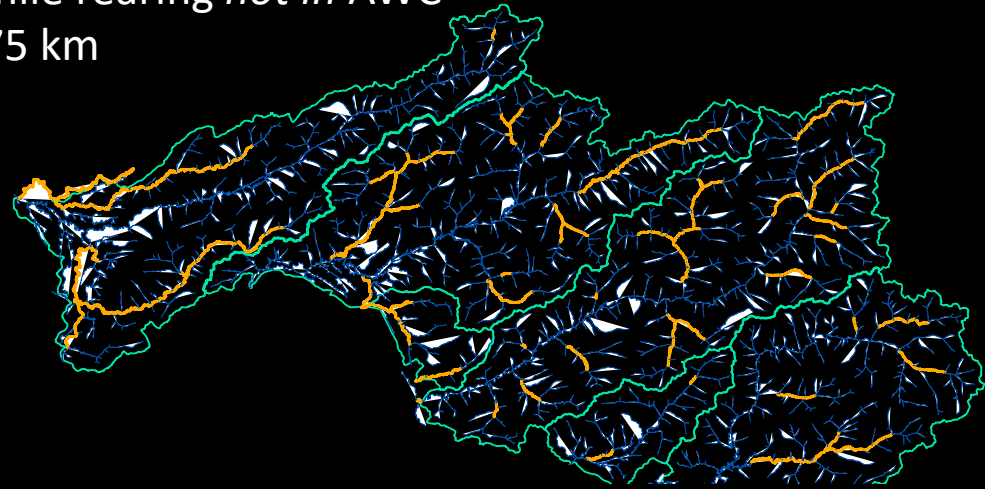
Y-K Habitat Potential: Applications

What "high" IP locations are outside the AWC?

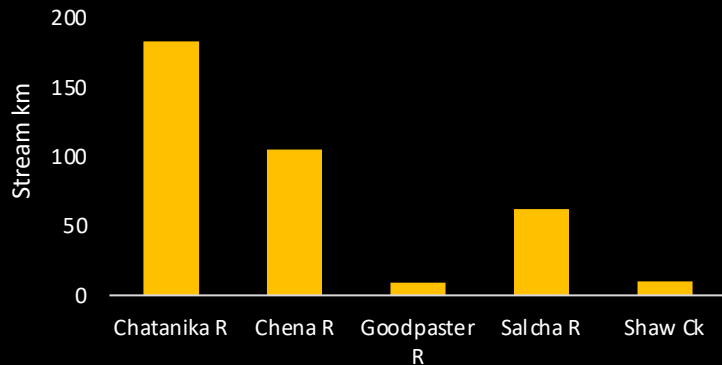
Juvenile rearing *not in* AWC



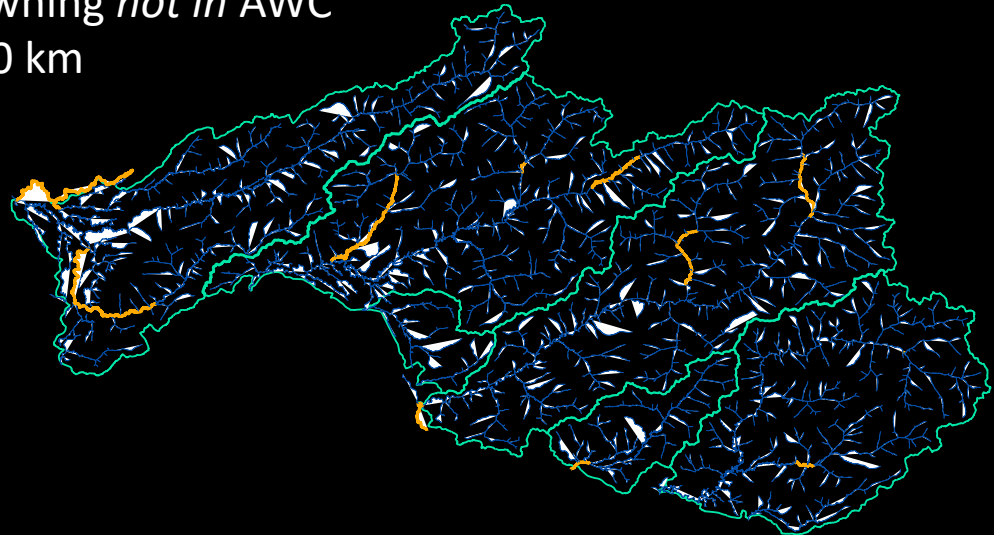
Juvenile rearing *not in* AWC
~1275 km



Spawning *not in* AWC



Spawning *not in* AWC
~370 km



Freshwater Habitat Potential for Chinook Salmon in the Yukon and Kuskokwim River Basins, Alaska

Project Goals

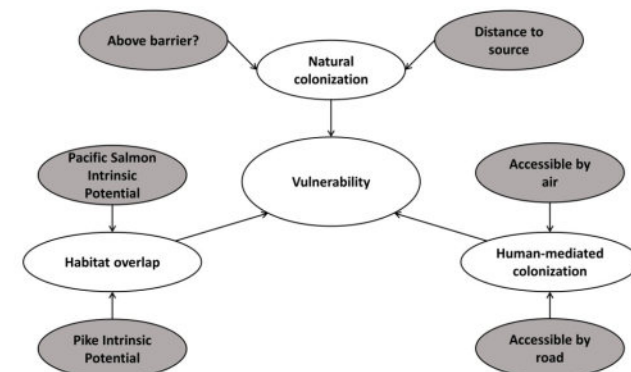
1. Develop synthetic stream networks for Y-K basins
2. Compile georeferenced juvenile & adult Chinook salmon distribution across Y-K basins
3. Produce maps of spawning & rearing habitat potential from modeled geomorphic attributes
- 4. Hold targeted stakeholder meetings to refine initial maps & models**
5. Develop a spatially explicit decision support tool

Stakeholder Feedback

Solicit input and expert opinion from stakeholders to fine-tune suitability curves and habitat potential models

Workshop Goals

- Present visualizations of results
- Develop initial conceptual models
- Explore potential management objectives



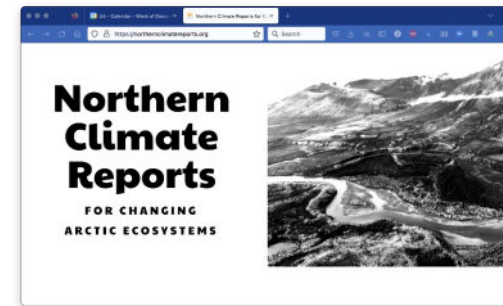
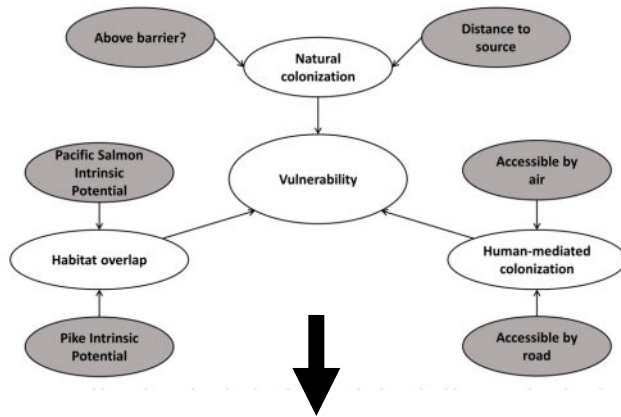
Freshwater Habitat Potential for Chinook Salmon in the Yukon and Kuskokwim River Basins, Alaska

Project Goals

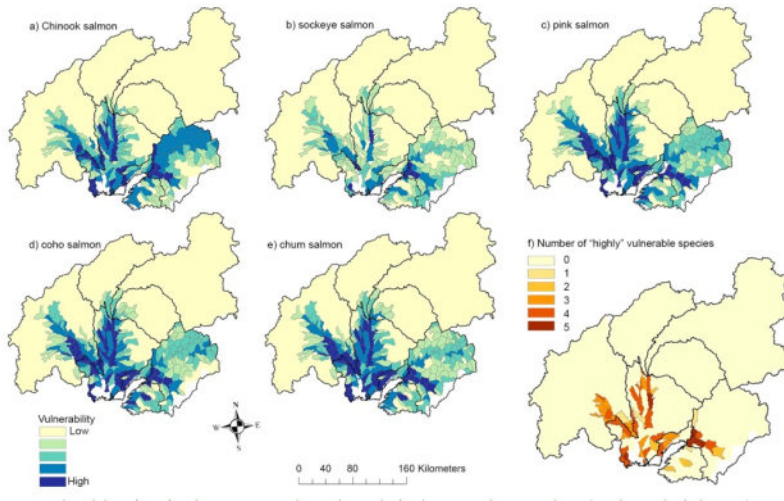
1. Develop synthetic stream networks for Y-K basins
2. Compile georeferenced juvenile & adult Chinook salmon distribution across Y-K basins
3. Produce maps of spawning & rearing habitat potential from modeled geomorphic attributes
4. Hold targeted stakeholder meetings to refine initial maps & models
5. **Develop a spatially explicit decision support tool**

Decision Support

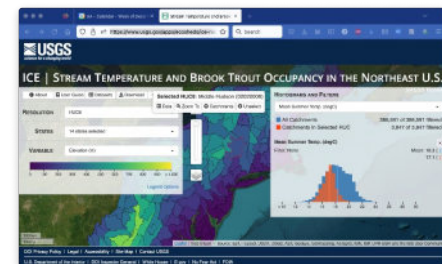
Develop a spatially-explicit decision-support tool to facilitate conservation and management of Chinook salmon in the Yukon and Kuskokwim river basins in Alaska.



<https://northernclimatereports.org/>



ArcGIS Online/Web Mapping



Jalbert et al. 2021

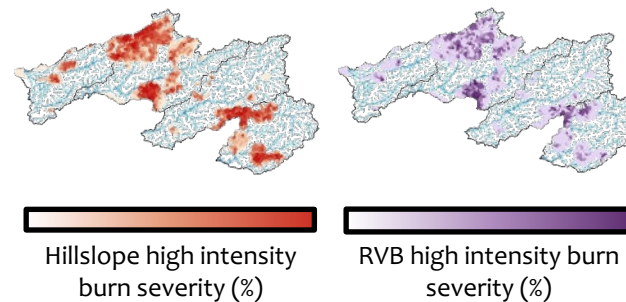
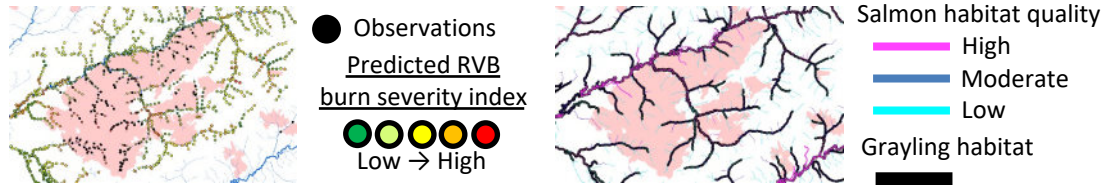
Ongoing/Future Work Incorporating Y-K IP Results

Boreal Fish and Fire (DoD – SERDP)

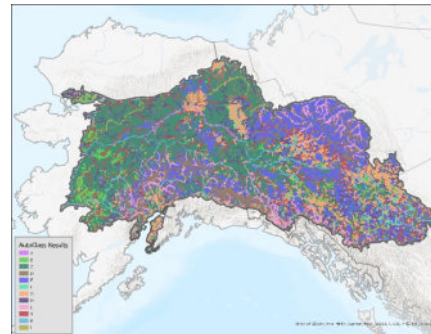
- Overlap between fish habitats and wildfire (Falke et al.)
- Growth potential of juvenile Chinook salmon under fire and climate scenarios (Schoen et al.)

Boreal Fish and Fire+ (AK CASC)

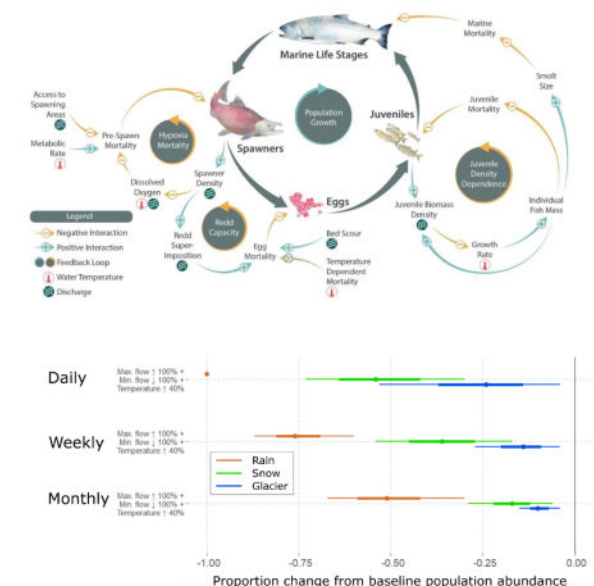
- Hydrologic Regimes
- Wildfire Impacts
- Stream Temperature
- Salmon Life Cycles (Shaftel)



Falke et al. *in prep*



Strohm et al. *in prep*



Bellmore et al. 2023

JAMMIN' SALMON



**Jeff Falke, U.S. Geological Survey
Alaska Cooperative Fish and Wildlife Research Unit
Institute of Arctic Biology &
College of Fisheries and Ocean Sciences
University of Alaska Fairbanks**
jfalke4@alaska.edu
(907) 474-6044



ALASKA